Part C: GAN-based Data Augmentation for CIFAR-10

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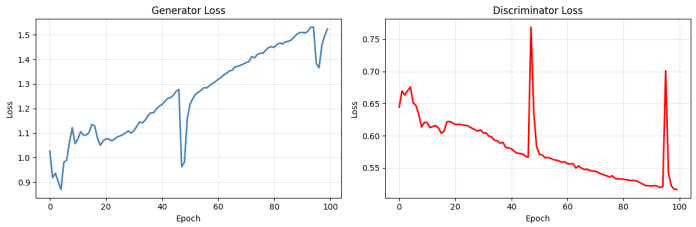
1. CGAN Training & Generated Samples

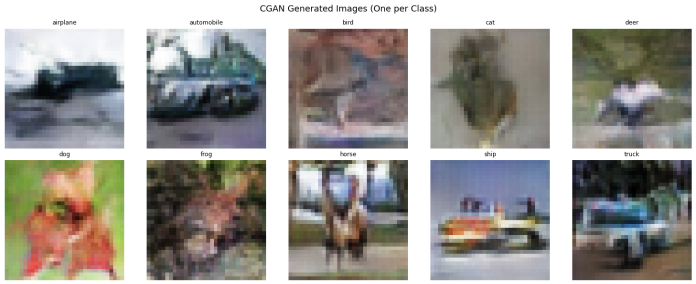
A Conditional GAN (CGAN) was trained for 100 epochs on the CIFAR-10 training set (50,000 images). The generator takes a 128-dim noise vector + one-hot class label and upsamples through Conv2DTranspose layers to produce 32x32x3 images. The discriminator receives an image concatenated with the class label and outputs a real/fake probability.

Both networks use Adam (lr=0.0002, beta_1=0.5) with binary cross-entropy loss. Label smoothing (real=0.9) is applied for training stability.

The loss curves show the generator loss increasing (~0.9 to ~1.5) while the discriminator loss decreases (~0.65 to ~0.53), indicating active adversarial dynamics without mode collapse.

Generated samples show recognisable structure for most classes. Vehicles (automobile, truck, ship) and animals with distinctive shapes (horse, frog) exhibit clear patterns. Classes with high intra-class variation (bird, cat) appear blurrier. Compared to VAE outputs from Part B, GAN images are notably sharper with more realistic textures.





2. Results

Model	Test Accuracy	Macro F1
Baseline	0.6402	0.65
GAN +10/class	0.6836	0.68
GAN +50/class	0.6670	0.68
GAN +100/class	0.6724	0.67

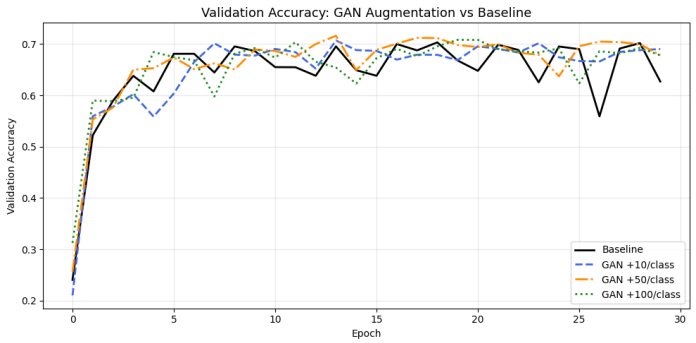
All GAN-augmented models outperform the baseline. The best result is +10 samples/class (68.36%), a +4.34% gain over the baseline (64.02%). Increasing to 50 or 100 per class yields slightly lower accuracy (66.70%, 67.24%).

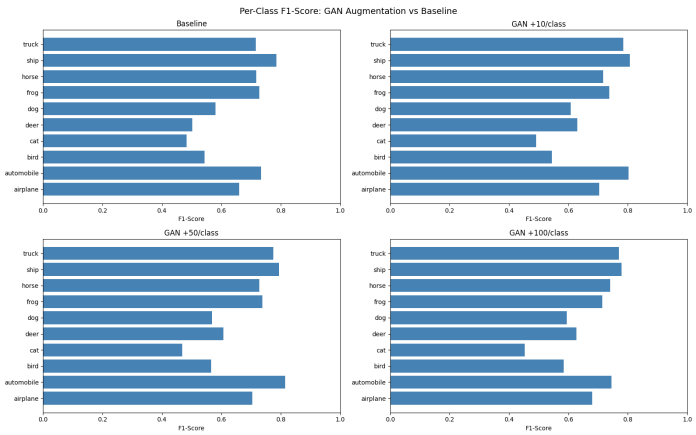
Validation accuracy curves show all models converging to the 0.65-0.72 range by epoch 30. Augmented models exhibit marginally smoother convergence.

Training loss curves are nearly identical across all configs, with augmented models showing slightly higher loss - a mild regularisation effect from the synthetic samples.

Per-class F1-scores show the most improvement in classes with distinct visual features: automobile (0.73->0.80), truck (0.72->0.78), deer (0.50->0.63). Difficult classes like cat (0.48->0.49) and bird (0.54->0.54) see minimal change due to high intra-class variability.

Confusion matrices confirm that GAN augmentation reduces severe misclassifications - deer correct predictions jump from 380 to 630 with +10/class. The cat-dog confusion persists across all configurations due to inherent visual similarity at 32x32 resolution.





3. Discussion: Why and Why Not There Is Improvement

3.1 Why GAN Augmentation Improves Accuracy

The GAN-generated images act as a regulariser by introducing additional training diversity. Unlike the VAE from Part B whose outputs are blurry pixel-wise averages, adversarial training produces sharper images with realistic local textures that lie closer to the true data manifold. This provides more useful gradient signal during CNN training.

Even adding just 10 samples per class (100 total, only 0.22% of the 45,000 training set) yields a meaningful +4.34% accuracy gain. This suggests GAN samples introduce novel variations (slightly different poses, backgrounds, colour shifts) that the CNN benefits from, improving generalisation to unseen test data. The improvement is most visible in classes where the GAN captures distinctive features well (vehicles, horse, frog).

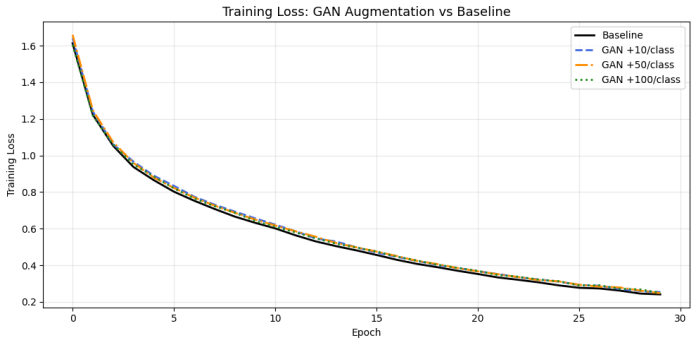
3.2 Why More Samples Do Not Always Help

Increasing from 10 to 50 or 100 per class does not improve accuracy further (68.36% -> 66.70% -> 67.24%). Three factors explain this:

1. Mode redundancy: The generator has limited mode coverage. More samples from the same generator increasingly repeat learned modes, adding redundancy rather than genuine diversity.

2. Distribution shift: As synthetic-to-real ratio grows, the CNN's training distribution shifts from the true test distribution. The CNN may overfit to GAN artefacts (characteristic blurriness, colour biases) absent in real test images.

3. Quality ceiling: For difficult classes (cat, bird), generated samples introduce more noise than signal. Cat F1-score actually drops from 0.49 to 0.45 at +100/class, showing that low-quality synthetic data can hurt specific class performance.



3.3 Comparison with VAE Augmentation (Part B)

GAN augmentation shows a clear advantage over VAE augmentation. The key difference is in the training objectives: VAEs optimise a pixel-level reconstruction loss that produces blurry outputs averaging over modes, while GANs use an adversarial loss that encourages sharp, realistic outputs. As a result, GAN-generated images provide more informative training signal to the CNN. However, neither approach achieves dramatic improvements because the original CIFAR-10 training set (5,000 images/class) is already large. Generative data augmentation is most impactful in low-data regimes - with 50,000 real samples already available, adding a few hundred synthetic ones provides only marginal gains.

4. Conclusion

A Conditional GAN was trained on CIFAR-10 and used for class-conditional data augmentation. The best configuration (GAN +10/class) achieved 68.36% test accuracy, a +4.34% improvement over the 64.02% baseline. Increasing generated samples to 50 or 100/class did not yield further gains due to mode redundancy and distribution shift. The GAN approach outperforms Part B's VAE augmentation, producing sharper synthetic images that better serve the downstream CNN classifier.